IGCSE 0607 Practice Questions — E5.5

Angles

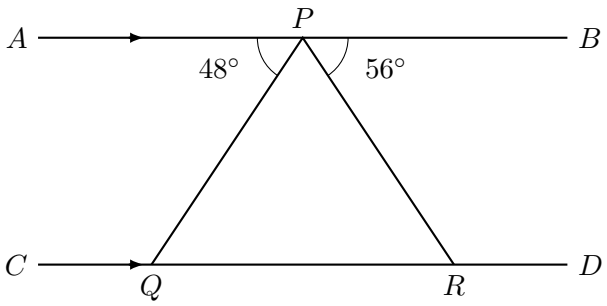
Total Questions: 15 — **Total Marks:** 92 Marks

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1 In the diagram, the line AB is parallel to the line CD.
P is a point on AB.
Q and R are points on CD.
The line PQ and the line PR are drawn.
Angle $APQ = 48°$ and angle $BPR = 56°$.



NOT TO SCALE

(a) Write down the value of angle PQR.

Angle $PQR =$ ..... ° **[1]**

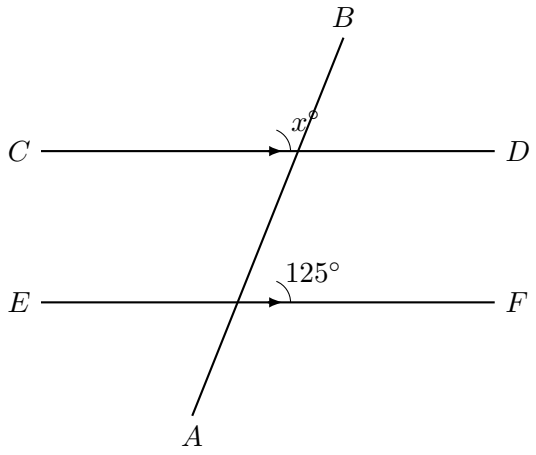
(b) Write down the value of angle PRQ.

Angle $PRQ =$ ..... ° **[1]**

(c) Work out the value of angle QPR.

Angle $QPR =$ ..... ° **[1]**

2 The diagram shows a straight line AB intersecting two parallel lines, CD and EF.



(a) Find the value of x.

$x =$ ..... **[1]**

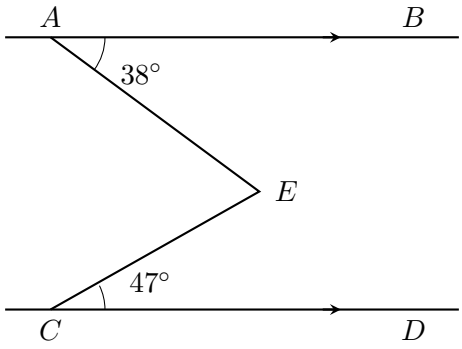
(b) Give a geometric reason for your answer to part (a).

..... **[1]**

(c) Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

3 In the diagram, the line AB is parallel to the line CD.
The point E lies between the two parallel lines.



NOT TO SCALE

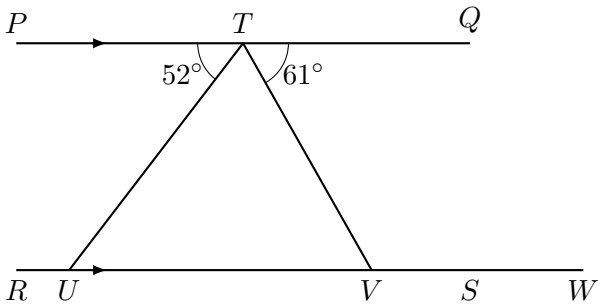
(a) Calculate the size of angle AEC.

Angle AEC = ... ° **[2]**

(b) A regular polygon has an exterior angle of 24°.
Work out the number of sides of this polygon.

..... **[2]**

4 In the diagram, the line PQ is parallel to the line RS.
The point T lies on PQ, and the points U and V lie on RS.
The line RS is extended to a point W.
Angle $PTU = 52°$ and angle $QTV = 61°$.



(a) Calculate angle TUV.

Angle TUV = ... ° **[1]**

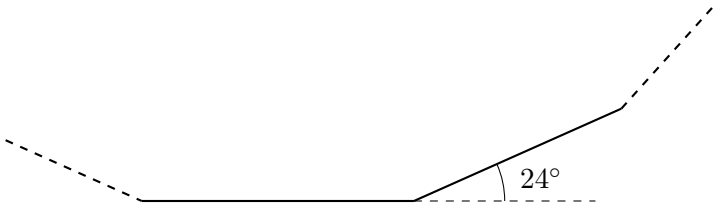
(b) Calculate angle UTV.

Angle UTV = ... ° **[2]**

(c) Calculate angle TVW.

Angle TVW = ... ° **[2]**

5 The diagram shows part of a regular polygon with n sides.
The exterior angle of this polygon is 24°.



(a) Calculate the value of n.

$n =$ ..... **[2]**

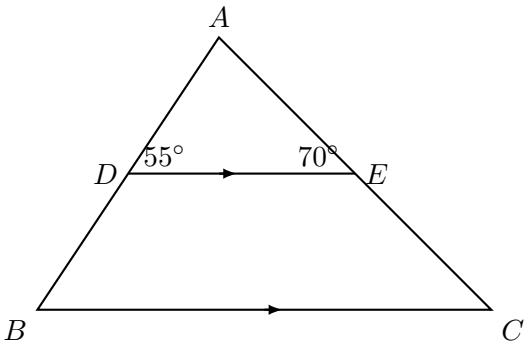
(b) Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

(c) A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

6 The diagram shows a triangle ABC with a line DE drawn parallel to BC.
D lies on AB and E lies on AC.
Angle $ADE = 55°$ and angle $AED = 70°$.



(a) Calculate angle DAE.

Angle $DAE =$ ..... ° **[2]**

(b) Calculate angle ABC.

Angle $ABC =$ ..... ° **[1]**

(c) Explain why triangle ADE is an isosceles triangle.

[2]

7 Answer the following questions about polygons and angles.

(a) An irregular polygon has n sides.
Three of the interior angles are each $140°$.
The remaining $(n - 3)$ interior angles are each $165°$.

Work out the value of n.

$n =$ ..... **[4]**

(b) In triangle ABC, the angles are $\angle A = a°$, $\angle B = b°$ and $\angle C = c°$.
It is given that $c - b = a$.

Prove that ABC is a right-angled triangle.

[2]

(c) Polygon P is a regular polygon with 12 sides.
Polygon Q is a regular polygon with k sides.
The exterior angle of Polygon Q is half the size of the exterior angle of Polygon P.
Polygon P and Polygon Q share a common edge AB.

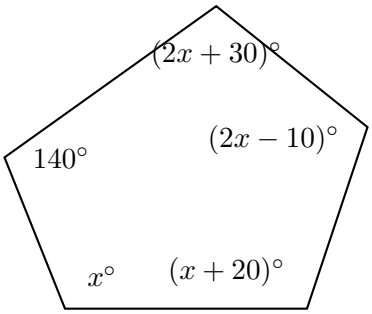
(i) Find the value of k.

$k =$ ..... **[2]**

(ii) Work out the size of the angle at vertex B that lies outside both polygons.

..... ° **[2]**

8 The diagram shows an irregular pentagon.
The interior angles of the pentagon are $x°$, $(x + 20)°$, $(2x - 10)°$, $(2x + 30)°$ and $140°$.



NOT TO SCALE

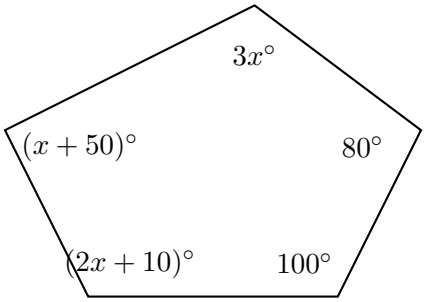
(a) Calculate the value of x.

$x =$ ..... **[3]**

(b) Find the size of the largest interior angle of the pentagon.

..... ° **[2]**

9 The diagram shows an irregular pentagon.



NOT TO SCALE

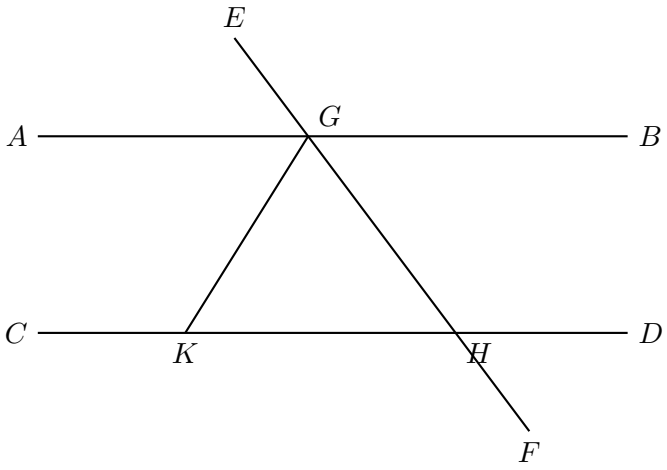
(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Another regular polygon has an interior angle that is 4 times the size of its exterior angle.
Calculate the number of sides of this polygon.

..... **[3]**

10 In the diagram, the line AB is parallel to the line CD.
The transversal EF intersects AB at G and CD at H.
The point K lies on CD.



(a) Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of x.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

(b) Angle $HGK = 35°$.
Calculate angle GKH.

..... ° **[3]**

11 **(a)** This question is about the interior and exterior angles of polygons.

(i) The interior angles of a hexagon are $x°$, $2x°$, $(x + 30)°$, $(2x + 10)°$, $(3x - 20)°$, and $160°$.
Work out the value of x.

$x =$ ..... **[3]**

(ii) A regular polygon has an interior angle that is 6.5 times its exterior angle.
Calculate the number of sides of this polygon.

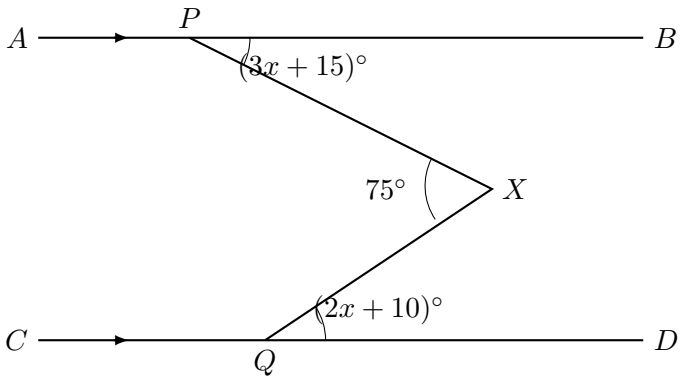
..... **[3]**

(iii) Two regular polygons, P and Q, have m sides and n sides respectively.
The sum of their exterior angles is $54°$.
The difference between their interior angles is $18°$.
Find the value of m and the value of n.

$m =$ ..... , $n =$ ..... **[5]**

12 **(a)** In the diagram, the line AB is parallel to the line CD.
X is a point between AB and CD.
The point P is on AB and the point Q is on CD.
Angle $BPX = (3x + 15)^\circ$, angle $DQX = (2x + 10)^\circ$ and angle $PXQ = 75^\circ$.

Find the value of x.



NOT TO SCALE

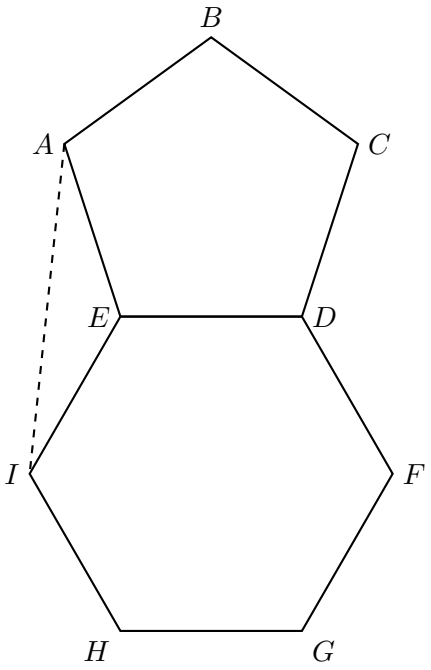
$x =$ ..... **[3]**

(b) Polygon P is a regular polygon with n sides.
Polygon Q is a regular polygon with $3n$ sides.
The size of an interior angle of polygon Q is 16° greater than the size of an interior angle of polygon P.

Calculate the value of n.

$n =$ ..... **[4]**

13 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge ED.



(a) Calculate angle CDE.

Angle $CDE =$ ..... ° **[2]**

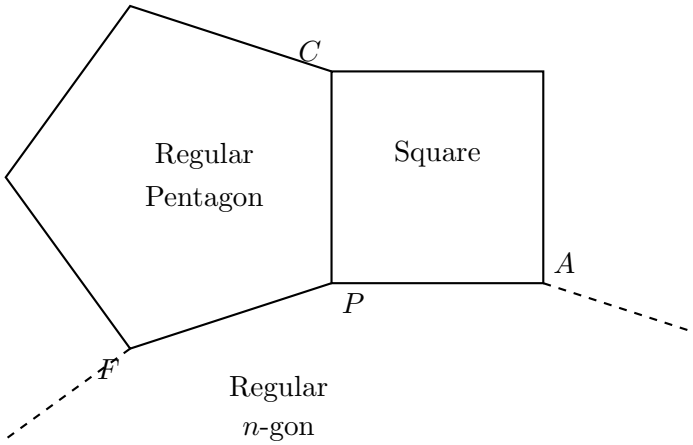
(b) Calculate angle AEI.

Angle $AEI =$ ..... ° **[2]**

(c) Calculate angle EAI.

Angle $EAI =$ ..... ° **[3]**

14 The diagram shows three regular polygons meeting at the point P.
The polygons are a square, a regular pentagon, and a regular polygon with n sides.
The square shares the side PC with the pentagon and the side PA with the n-gon.
The pentagon shares the side PF with the n-gon.
They meet with no gaps and do not overlap.



(a) Calculate the interior angle of a regular pentagon.

..... ° **[2]**

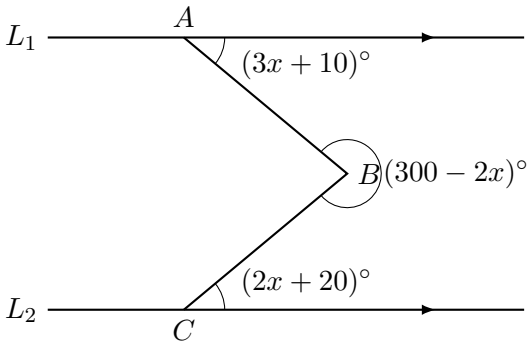
(b) Find the value of n.

$n =$ ..... **[3]**

(c) The vertices A and F of the n-gon are joined by a straight line.
Calculate angle PFA.

..... ° **[2]**

15 The diagram shows two parallel lines, L_1 and L_2.
The point A lies on L_1 and the point C lies on L_2.
The point B lies between the two parallel lines.
The acute angle between L_1 and AB is $(3x + 10)°$.
The acute angle between L_2 and CB is $(2x + 20)°$.
The reflex angle ABC is $(300 - 2x)°$.



(a) Show that $x = 10$.

[4]

(b) The points A, B, and C form a triangle ABC.
Given that $AB = BC$, calculate the angle BCA.

Angle BCA = ... ° **[3]**